

ccint — Progress Brief

Canada Cyber Social Intelligence: design rationale, build, and where it stands

22 September 2026

The question, and what changed about it

The project began as a longitudinal study: *how does Canada-related cybersecurity discussion change over time on social media?* A collector runs continuously against Bluesky, posts are labelled under versioned schemes, and change is tested statistically.

The first full analysis returned a negative result, and returned it with enough rigour to be worth something. The largest topic movement in the 30-day window, `data_breach` at +22%, has $p_{\text{FWER}} = 0.734$ under an author-level circular-shift null that preserves each author’s volume, topic mix and burstiness while destroying calendar alignment. Empirical power analysis — measured by author-pool replication rather than assumed — puts the requirement at roughly $8\times$ the current volume.

That result is only interesting if one knows what produced it. Three explanations were open, and nothing in the system could separate them, because every figure it produced came from labels it had produced itself. The work of the last phase was to build the measurements that can separate them. They do, and the answer was not the expected one: **the shortfall is not in the statistics and not mainly in the labelling — it is that the platform carries almost no human discussion of these events at all.**

Why the system is built this way

Four commitments shaped every design decision. Each exists to prevent a specific failure that would be invisible after the fact.

Bluesky, and one platform only. An open, documented API with usable historical search and terms that permit research collection; X and Reddit were excluded on access grounds rather than merit. The consequence is stated rather than hidden: this is a convenience sample, and the project’s job became partly to measure what it can show.

Labels are versioned and never overwritten. `post_labels` is keyed (`post_id`, `label_version`) and rows are only ever inserted. Four labelling schemes currently coexist over the same corpus and can be diffed row by row. Without this, every improvement to the labeller would silently rewrite history, and a change in topic distribution could never be attributed to the world rather than to the method.

Raw data is kept verbatim; every number carries provenance. `raw_payload` is stored unmodified, and every run and model label records what produced it — query version, label version, method, and for model labels the SHA-256 of the exact prompt text. The last was added recently, after noticing that a model labeller’s output is entirely determined by its prompt: “labels are never overwritten” had been preserving the labels while losing the conditions that made them.

System failure must be distinguishable from social change. Every collection path writes to `collection_runs` with mode, status, window and counts. A day with half the usual volume must be attributable to either the world or a token expiry, and only bookkeeping makes that possible. This principle held under test: a 42-run token outage is visible in the record as 42 `partial` runs rather than as a drop in Canadian cyber discussion.

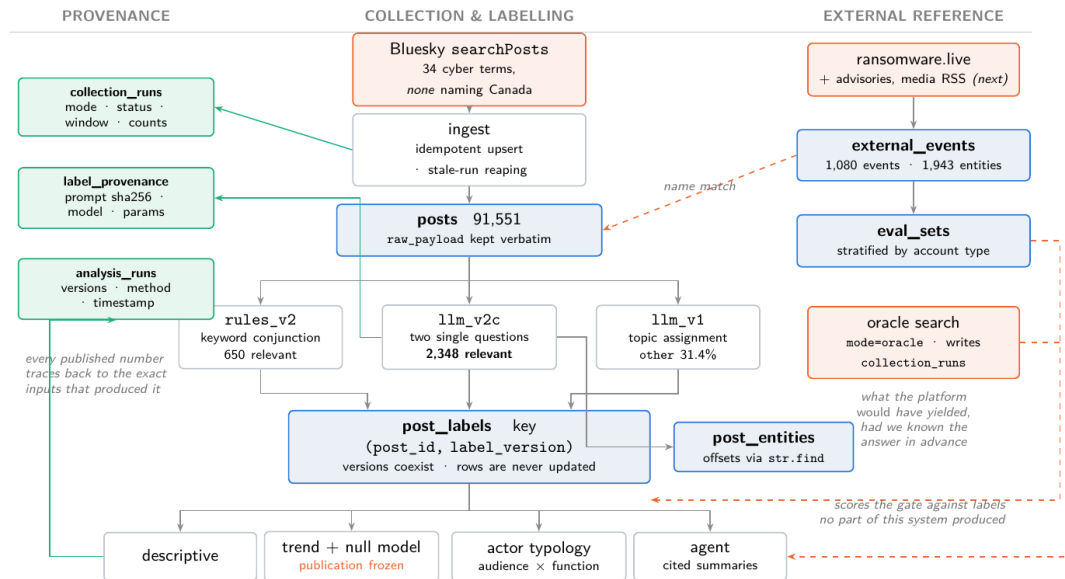


Figure 1: System architecture. Blue is durable state, orange is the world outside the system, green is the provenance rail that every path must write to. The external-reference column on the right is the most recent addition and the one that changed the project’s conclusions.

Underlying all four: **measure before improving**. Repeatedly the intuitive next step turned out to be aimed at the wrong quantity, and only measurement revealed it — tuning the trend score would have been wasted, relaxing the relevance rules helps but not for the population that seemed to need it, and the most recent prompt improvement introduced a regression the evaluation set certified as correct.

How it was built

Collection runs in two modes against the fixed term list — a day-by-day backfill and an hourly incremental — with idempotent ingestion, stale-run reaping and client-timestamp repair. The corpus stands at 91,551 posts over 22 August – 22 September 2026.

Labelling proceeded in four generations, all retained. **rules_v1** and **rules_v2** are lexicon-based; **llm_v1** re-decides topic only, inheriting the relevance verdict verbatim so the two changes stay separately attributable; **llm_v2c** replaces the relevance gate. Analysis covers descriptive statistics, permutation testing against two null models, behavioural broadcast detection with a collection-lag control, and a two-axis actor typology separating measured audience from content-judged function. The most recent phase added the right-hand column of Figure 1: an external registry, evaluation sets built from it, and an oracle collection mode that asks what the platform *would* have yielded had the answer been known in advance.

Where it stands

The collection boundary was never measured

The 34 search terms came from a specification naming four examples followed by “etc.”; the remaining 26 were supplied during implementation from domain intuition, never derived from data and never revised. Reconstructing per-term productivity offline over the stored corpus:

Search term	Posts retrieved	Judged relevant
cybersecurity	22,498	200
ransomware	8,881	105
data breach	1,939	90
CVE	12,502	3
11 further terms	—	0

CVE accounts for 13.7% of everything collected and returns three relevant posts. `cybersecurity` alone accounts for 30.8% of all relevant posts: the corpus time series is, to first order, one search term’s time series, and nothing monitors it. None of the 34 terms names Canada, so any Canadian cyber post not phrased with one of them is unreachable by construction.

Replacing the relevance gate

The old gate was a conjunction of two narrow lexicons, so recall was bounded by their intersection: 650 of 91,551 posts passed, and the 89,714 rejected had never been examined by anything semantic. The replacement asks a local model two *independent* single questions rather than one conjunction — a single combined prompt causes a 4B model to collapse the conjunction onto the side it knows better, admitting global threat-intelligence noise at 6.8% while the true Canadian rate is 1.1%.

Relevance gate	recall @ registry	specificity @ registry	corpus relevant
rules_v2 — keyword conjunction	46.5%	99.8%	650
llm_v2c — two questions + validation	68.3%	99.4%	2,348

Of the 1,799 posts newly admitted, 84.0% had been rejected on the Canadian side of the conjunction. Two findings from this phase are worth carrying forward as method rather than as numbers. First, entity character offsets are computed with `str.find` rather than requested from the model, on the narrow grounds that small models miscount them — and this turned into a free hallucination detector, since an entity that cannot be located in the post was invented. Nearly half of extracted `place` entities failed it. Second, the residual error that remains is not a prompt or capability problem: the posts still missed contain no country marker at all (“New post from #Qilin: Coldfish Seafood”), so closing it requires a gazetteer of Canadian organisations rather than a better model.

The external reference, and the central result

Until this phase every precision and recall figure traced to a single author who wrote the rules, the prompts, and the hand-coded evaluation set. A reported $\kappa = 0.870$ measured agreement under those conditions, not independence. Ingesting a public ransomware victim registry (1,080 events, 1,943 entities) supplies labels no part of the system produced.

For every Canadian event in the window, the victim organisation’s name was also searched directly on the platform, with no cybersecurity term attached, and every hit passed through the cyber judgement before counting:

Canadian ransomware events, 22 Aug – 22 Sep 2026	Events	Share
Listed in the registry	27	—
At least one cyber-related post exists on Bluesky	26	96%
At least one such post was collected by ccint	26	96%
At least one came from a non-feed account	2	7%

The 144 relevant posts come from 15 accounts. Twelve are automated threat-intelligence feeds; the three that are not are Canadian news organisations contributing five posts between them, all about a single event. Individual users contributed nothing; 94.9% of the posts have no likes.

What looks like coverage is syndication. The collector is not the bottleneck here — it captured 26 of the 26 events the platform carried. No ransomware trend is detectable because there is almost no ransomware *discussion* to detect: the corpus is a near-complete mirror of several leak-site feeds. That also reframes the power analysis — 8× of this corpus is 8× as much syndication, so volume is the wrong lever.

The one exception is instructive. The Commission de la construction du Québec, listed by the Qilin group with roughly 350,000 records taken, drew genuine Canadian media coverage in French and English. Four of the five posts are in the corpus; the missing one, a radio interview with the CCQ’s chief executive, contains no term from the 34-term list because *protection préventive* is in no lexicon. One post, one missing phrase, a real event — the collection-boundary gap at human scale.

A methodological result about evaluation

The prompt revision that fixed recall also generalised “a .ca domain counts” into “a domain counts”, admitting `trustedsec.com`, the UK’s NCSC and `ia.cr` — the Costa Rican TLD — and inflating the corpus from 906 to 3,528 posts, a third of them spurious. The external negative set reported 99.0% specificity throughout, because it is built from formulaic English feed posts about foreign victims: no URLs, no French, and so none of the model’s actual failure modes. **An external label removes self-reference but does not confer representativeness, and a non-representative evaluation set does not merely fail to catch a regression — it certifies one.**

What the project is, and what comes next

The original framing — trend analysis of Canadian cyber discourse — is not available at this volume for this event class, and adding data will not make it so. What is available, and appears to be unreported in the surveyed literature, is a sensor that states its own blind spots numerically. Published systems report how many events they detected (84 of 151 in the strongest comparable work); none reports how many events had no platform presence at all, because establishing that requires the per-event oracle search used here, affordable only when the event count is small. This project’s greatest liability — a corpus of a few thousand relevant posts — is precisely what makes the denominator measurable.

Immediate priorities, in order:

1. **Widen the registry beyond ransomware** — Cyber Centre advisories, CISA KEV, Canadian media RSS. Every figure above is conditioned on a source seeing only leak-site victims: the narrowest limitation in the work.
2. **Feed the registry back into the gate.** The residual failure is missing knowledge of Canadian organisation names; the registry is a list of exactly those names.
3. **Add a Canada-side query stream** as an additional OR stream, not a filter — to reach the population the registry cannot see.
4. **Have an independent annotator label the 200-post sample** already drawn and left blank. It is the one step that cannot be automated, and the one every other number rests on.

Corpus 91,551 posts, 22 Aug – 22 Sep 2026; registry retrieved 22 Sep 2026; four coexisting label versions; 243 tests passing. Methodology in the project wrap-up; coverage measurements in the iteration report.